

# Exploring Trust and Transparency in Retrieval-Augmented Generation for Domain Experts

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## Abstract

The increasing adoption of Retrieval-Augmented Generation (RAG) systems in domain-specific applications raises critical challenges around trust and transparency, particularly for domain experts who possess deep subject knowledge but limited technical familiarity with AI systems. This paper investigates the design and evaluation of a user-centric RAG-based application tailored for financial equity research, incorporating features such as confidence indicators, source transparency, and user control mechanisms. Through a user study with 50 financial professionals, we found that while confidence scores provided useful context, they alone did not significantly enhance trust. In contrast, transparency features, including source attribution and highlighting document sections used in responses, substantially improved trust and user understanding. Participants also expressed a strong preference for greater control over source documents, such as filtering trusted reports and interacting with specific data. This work provides actionable insights for designing RAG systems that support trust, transparency, and control in high-stakes decision-making domains.

## CCS Concepts

- **Human-centered computing** → **Natural language interfaces;**
- **Computing methodologies** → **Natural language processing;**
- Natural language generation.**

## Keywords

Retrieval Augmented Generation, Trust, Transparency, Control

## ACM Reference Format:

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## 1 Introduction

The rapid advancement of *Large Language Models (LLMs)* has revolutionized natural language processing (NLP), enabling applications in automated content generation, translation, and question-answering (QA) systems. However, the adoption of these systems in *high-stakes domains* such as finance, law, and healthcare has introduced significant challenges, stemming not only from concerns about the *accuracy* and *reliability* of generated responses but also from the need to establish *trust* among domain experts who rely on these systems for decision-making. The consequences of errors in these contexts can be severe, ranging from financial losses to legal or medical misjudgments [8]. Addressing these challenges requires AI systems that prioritize trustworthiness, transparency, and user engagement.

*Retrieval-Augmented Generation (RAG)* systems present a promising solution by integrating LLMs with external retrieval mechanisms. This hybrid architecture combines the *parametric knowledge* of pre-trained models with *real-time retrieval* of domain-specific information from external knowledge bases [15, 21]. By grounding responses in verifiable information, RAG systems enhance factual accuracy and contextual relevance, making them particularly suitable for specialized domains. Despite these advantages, their real-world deployment often encounters barriers rooted in user trust, especially when the intended users possess deep domain expertise but limited familiarity with AI systems [24].

For domain experts, *trust* is a multifaceted construct that extends beyond the system's technical performance. It encompasses perceptions of *reliability*, *transparency*, *safety*, and *user control* [3]. Among these dimensions, *transparency* plays a critical role in enabling users to understand how the system generates responses, evaluate its limitations, and build confidence in its outputs [18]. Transparency can be operationalized through features such as *source attribution*, *confidence scores*, and *explanations*, which allow users to trace information back to its origin. However, transparency without corresponding mechanisms for *user control*—such as the ability to filter sources or explore supporting evidence—can lead to frustration and undermine trust [18, 27].

While prior work on RAG systems has explored technical improvements to retrieval accuracy and response generation [30], limited attention has been given to *human-centered design principles* that foster trust and transparency for *non-technical domain experts*. Existing studies have predominantly focused on technical users or general-purpose applications, leaving a gap in understanding how to design RAG systems that align with the needs of expert users in *high-stakes decision-making contexts*. To explore this gap,

we present the design and preliminary evaluation of an *Equity Research Assistant*, a RAG-based application tailored for the financial domain. The system allows users to query various stocks using expert analysts' reports, and incorporates features designed to enhance trust: confidence levels, source transparency, and user control mechanisms.

We conducted a user study with 50 participants from the financial domain to investigate how these design features influence trust, perceived transparency, and user engagement. Unlike prior studies, our work focuses on the interplay between trust and transparency in the context of domain-specific, semi-structured documents such as financial analyst reports. This paper makes the following contributions:

- *A user-centered RAG design approach* tailored to the needs and expertise of domain experts in a financial context.
- *Empirical evaluation of transparency-enhancing features* and their impact on user trust and understanding.
- *Actionable design recommendations* for developing trustworthy RAG systems in knowledge-intensive domains.

By addressing the intersection of trust, transparency, and usability, this research contributes to a broader understanding of how AI systems can be integrated into specialized domains while maintaining reliability and user confidence.

## 2 Related Work

Our work is informed by research on user trust, transparency, and control in AI systems, particularly concerning LLMs and their application in Retrieval-Augmented Generation (RAG). We explore these principles in RAG systems designed for domain experts.

**Trust, Transparency, and Control in AI:** Effective AI decision-making requires users to understand system limitations and build appropriate trust. Transparency and explainability are crucial for enabling comprehension of AI capabilities, reasoning, and limitations [1, 2, 19, 28]. However, transparency can lead to over-reliance on AI, even when erroneous [5], highlighting the need for calibrated trust. Explainability effectiveness is also user-dependent; domain expertise influences how users engage with explanations. Users often prefer "why" explanations and collaborative feedback opportunities over technical details [13, 22]. Thus, tailoring explanations to user needs is critical for trust and usability.

**Trust, Transparency, and Control in LLMs:** The use of LLMs introduces challenges for trust, transparency, and control. Uncertainty expression impacts user reliance, with first-person framing proving more effective than generalized statements [12]. Source attribution strengthens trust by enabling verification of information origins [11, 20]. Trust in LLMs encompasses confidence in the model, application, provider, and specific output types [18], requiring diverse transparency approaches. Control, such as interactive workflow chaining, enhances transparency, controllability, and user collaboration [26]. While some studies consider context within control [16], transparency within LLM applications remains under-explored.

**RAG and Applications:** RAG systems combine LLMs with external retrieval to improve factual accuracy by leveraging domain-specific information [15, 17], particularly useful in high-stakes

domains like finance [6, 31]. Trustworthiness in RAG includes factuality, robustness, fairness, transparency, accountability, and privacy [30]. Transparency is categorized into retrieval (how documents are selected) and information integration (how retrieved information is used). Research on LLM trustworthiness considers dimensions such as truthfulness, safety, fairness, robustness, privacy, and ethics [10]. While previous studies focus on technical aspects, we assess trust and transparency in RAG from a domain-user perspective, emphasizing confidence levels, contextual highlights, and user control (e.g., branching queries) within a financial context, aiming to improve user trust and usability in these high-stakes applications.

## 3 Equity Research Assistant Using RAG

This work investigates the user-centered design of a Retrieval-Augmented Generation (RAG) system tailored for high-stakes domains where trust and transparency are paramount. Our focus is the financial analysis domain, exemplified through a prototype, the Equity Research Assistant, which enables domain users to query multiple equity reports.

The prototype offers a streamlined interface where users can select specific equities, ask customized or pre-defined queries, and refer to past queries (Figure 1a). Pre-defined queries are crafted from commonly asked questions by financial analysts, facilitating usability for novice users. The system is currently designed to utilize analysts' reports for responses.

Our design integrates four guiding principles—trust, transparency, user control, and feedback—to meet the needs of domain users who may lack a technical understanding of AI mechanisms. The following subsections elaborate on key features informed by these principles.

### 3.1 Source Attribution and Relevance:

Financial equity analysis often requires considering diverse expert opinions across sources. Unlike conventional RAG systems, which synthesize a single response from multiple sources, our system provides **source-specific** answers. Each response is presented alongside a **confidence level** categorized as *LOW*, *MEDIUM*, *HIGH* to help users assess reliability (Figure 1b).

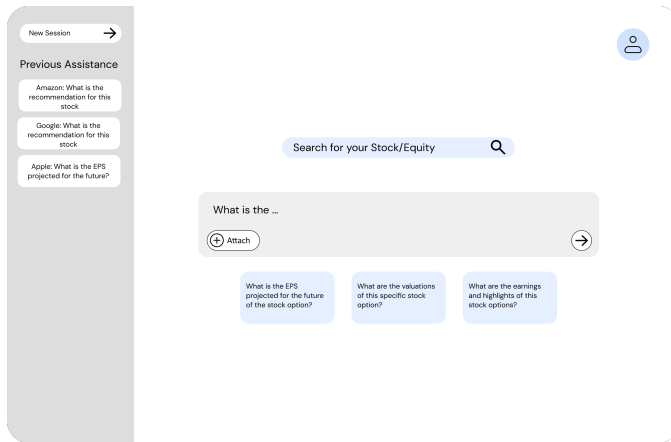
Confidence levels are computed based on two factors:

- **Relevance:** Derived from retrieval scores of document sections stored in a vector database.
- **Factuality:** Assessed through hallucination detection, comparing the generated response to its source context [4].

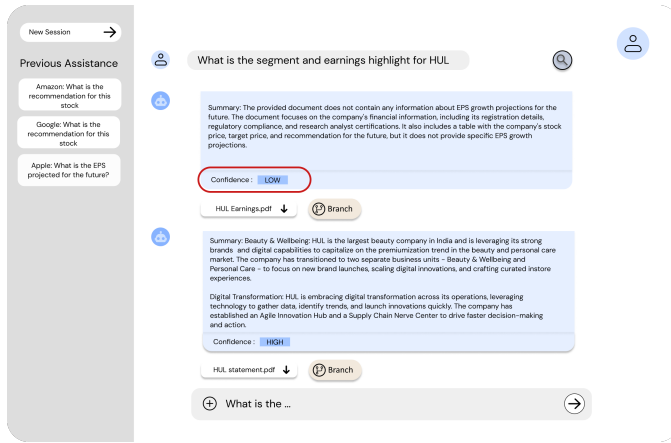
The numeric values between 0 and 1 are categorized into three levels and presented to the user. This approach supports trust calibration by encouraging users to critically evaluate responses. Additionally, a **meta-summary** synthesizes areas of consensus and divergence across sources, promoting informed decision-making. For instance, a low-confidence response signals users to prioritize higher-confidence outputs or explore the underlying source.

### 3.2 Transparency Through Highlighted Text:

To foster transparency, we highlight the specific sections of source documents used to generate each response [23]. Links accompanying each response direct users to the respective source PDFs, where



(a) RAG on Equities with pre-defined and reference queries



(b) Answers from different sources with confidence level

Figure 1: Equity Research Assistant

relevant sections are visually highlighted (Figure 2). This feature ensures:

- **Traceability:** enabling users to verify the provenance of the information,
- **Intelligibility:** helping users understand the rationale behind the generated responses.

By enabling users to explore the evidence directly, this design bolsters trust and enables understanding.

### 3.3 User Control Through Branching:

To accommodate diverse user workflows, we introduced a feature called branching, allowing users to manage multiple streams of inquiry. In our system, the user query is answered from multiple sources. However, users may want the ability to select a single source document and ask questions based solely on that document. This allows for deeper exploration of individual financial reports, which can be particularly useful when the user wants to focus on one source rather than aggregating information from multiple reports. The user can switch between getting responses from all sources, or can select a specific source that starts a separate thread.

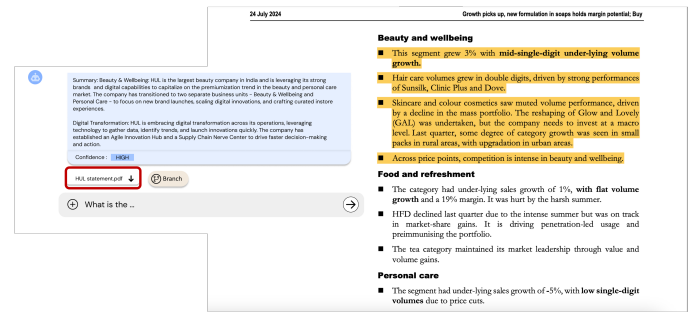


Figure 2: Highlight sections of the source document that were used to generate the answer

Each branch opens in a separate tab, preserving the context and sequence of the inquiry (Figure 3a). Key benefits include:

- **Focused Exploration:** Users can isolate and examine specific data without conflicting inputs.
- **Flexible Workflows:** Parallel branches allow simultaneous exploration of multiple sources or questions.

This feature enhances user agency, enabling them to tailor the system to their specific needs.

### 3.4 Intuitive Feedback Mechanisms:

An intuitive feedback mechanism enables users to evaluate system responses seamlessly [7]. The interface supports both categorical and qualitative feedback (Figure 3b): i) **Categorical Feedback:** Users rate responses based on predefined criteria, ii) **Qualitative Feedback:** Users provide detailed comments in natural language, offering context-rich insights. Aligned with principles of mixed-initiative systems [9], this design encourages human-machine collaboration. Feedback is instrumental in refining the system, improving retrieval, and tailoring response generation to user needs [14, 29].

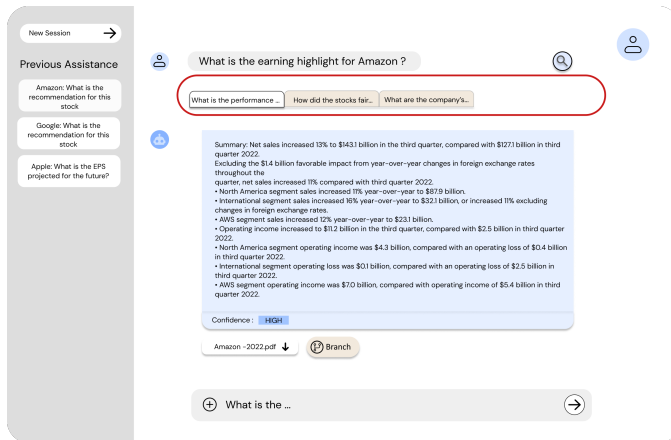
## 4 User Study

We conducted a study to evaluate how confidence levels, source attribution, and text highlighting impact user trust and transparency in a RAG system for financial analysis. The study explored preferences for control and feedback, informing design principles for high-stakes domains. Our study addressed the following questions:

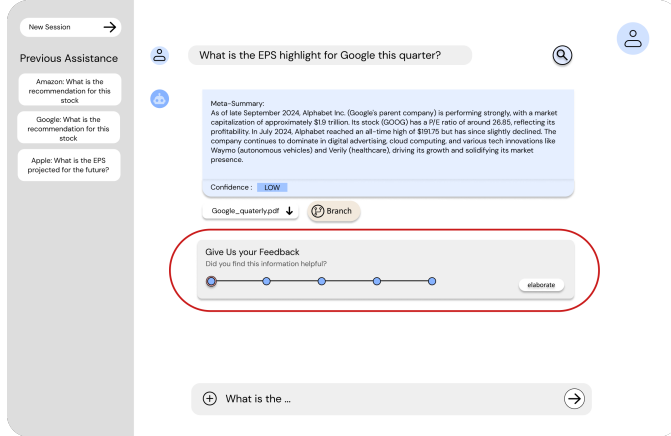
- **RQ1:** How do confidence levels and text highlighting affect user trust?
- **RQ2:** How does text highlighting impact understanding compared to confidence levels alone?
- **RQ3:** What user preferences exist for controls and additional functionality to improve usability?

### 4.1 Study Design

**Participants:** We recruited 50 participants through group messaging platforms such as LinkedIn and WhatsApp. Eligibility criteria included either a bachelor's degree in commerce or experience in stock trading or analysis. Participants' demographics and expertise were as follows:



(a) Branching leads to a new tab with responses from a specific document



(b) Collect feedback from users

Figure 3: User control and Feedback

- **Age Distribution:** 34 participants were aged 40–55, 10 were aged 25–40, and 6 were aged 20–25.
- **Financial Expertise:** 47 participants reported knowledge or experience in equities and financial markets.
- **AI Familiarity:** Participants rated their familiarity with generative AI systems (e.g., have used ChatGPT) on a 1–5 scale (1 = no experience, 5 = expert). The mean AI familiarity score was 2.64 (SD = 1.12), with 8 participants reporting no familiarity (score = 1), 26 with some familiarity (2 and 3), and 16 with moderate experience (4 and 5). We categorize these users as *High-domain, Low-AI* users [13]. They do not possess understanding of the AI system but only use it.

**Tasks and Conditions:** Participants interacted with the Equity Research Assistant in two conditions:

- **Confidence Levels:** The system displayed confidence levels for each response alongside source attribution.
- **Confidence Levels and Text Highlighting:** The system highlighted specific sections of source documents used to

generate responses, in addition to confidence levels and source attribution.

Each participant completed six tasks, with four tasks divided equally across the two conditions:

- **Task 1:** Participants asked a predefined query (e.g., What is the recommendation for stock X?). Stock X was selected from a predefined list of equities, ensuring the system had analyst reports available for generating responses. Responses included high-confidence levels, providing a straightforward evaluation of system outputs.
- **Task 2:** Participants asked another predefined query (e.g., What is the projected EPS for stock Z?). Stock Z was chosen from a predefined list. Responses included low-confidence levels, prompting participants to verify outputs using source documents.
- **Task 3:** Participants repeated Task 1, but with highlighted document sections for additional transparency.
- **Task 4:** Participants repeated Task 2, again with highlighted sections to aid understanding of low-confidence outputs.
- **Task 5:** Participants chose branching on any one of the source documents, and could ask any open-ended question (e.g., Why is the projection low?, What are the future plans?).
- **Task 6:** Participants were provided instructions to use the feedback capability and provide qualitative or quantitative inputs on the responses generated for any of the previous tasks.

Participants did not generate their own questions for the first 4 tasks, but followed a set of predefined queries. Predefined queries were selected to ensure consistency across participants and enable controlled comparisons between conditions. Tasks 1 and 2, and Task 3 and 4 simulate common equity research queries typically used by financial analysts when evaluating investment options. Task 5 and 6 engages users in providing feedback and having control. Hence, in for these tasks, the participants could generate their own questions with some examples and instructions provided.

**Procedure:** Before using the system, participants completed a baseline trust survey to assess their initial trust in AI-based system. The survey included Likert-scale questions (1–5) evaluating perceived trust by answering the question, *How much would you trust in an AI system that can assist you as an Equity Research Assistant?* This served as a reference point for comparing changes in trust after interacting with the system. Participants then used the system under both conditions. After each condition they rated trust and understanding on 1–5 Likert scale (i.e., after completing Task 1 and 2, and Task 3 and 4). Finally, participants answered open-ended questions on preferences for branching, feedback, and additional features. We opted to begin with the *Confidence Levels* condition to establish a baseline understanding of the system’s basic functionality before introducing the more complex *Confidence Levels and Text Highlighting* condition. This staged introduction aligns with a typical user onboarding experience, where new users would first learn the basic functions before exploring advanced features

## 4.2 Results and Insights

We answer RQ1 and RQ2 using quantitative data and RQ3 through open coding of qualitative data.

**RQ1: How do confidence levels and text highlighting affect user trust?:** We used Post-hoc Wilcoxon signed-rank tests to compare trust under the three conditions. Presenting Confidence levels with source attribution alone did not significantly increase trust ( $M = 3.00$ ,  $SD = 0.93$ ) compared to baseline trust ( $M = 2.76$ ,  $SD = 1.08$ ). The comparison did not reveal a statistically significant difference ( $p = 0.080$ ), suggesting that the inclusion of confidence levels alone did not notably influence trust. See Figure 4a for the distribution of user responses to, *How much do you trust the output/response of the system?*

Next, we compared the trust scores in the condition with confidence levels and text highlighting. We found a statistically significant increase in trust ( $M = 3.14$ ,  $SD = 1.26$ ) in comparison to the condition with confidence levels ( $p = 0.0036$ ), and to the baseline trust level ( $p = 0.0029$ ). After applying the Holm-Bonferroni correction for multiple comparisons, the adjusted p-values were  $p = 0.0801$  (baseline and confidence level),  $p = 0.0086$  (confidence level and text highlighting), and  $p = 0.0086$  (baseline and text highlighting), respectively.

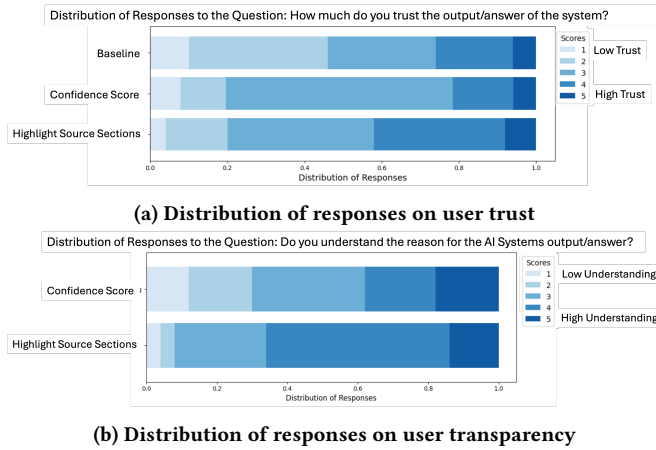


Figure 4: User responses on trust and transparency

**RQ2: How does text highlighting impact understanding?:** Text highlighting significantly increased perceived understanding ( $M = 3.68$ ,  $SD = 0.91$ ,  $p < 0.01$ ) compared to confidence levels alone ( $M = 3.14$ ,  $SD = 1.26$ ). See Figure 4b for the distribution of user responses to the question, *Do you understand the reason for the AI system's output/response?*. We conducted a Wilcoxon signed-rank test to compare participants' perceived understanding. The results revealed a statistically significant difference under the two conditions ( $p = 0.0021$ ). Thus, highlighting relevant sections of the documents enabled participants trace the reasoning behind the system's answers more effectively than when only confidence levels were presented.

**RQ3: What controls or functionality do domain users require from a RAG system?** This data is based on responses to

open-ended questions gathered after the study on their preferences to use *branching*, and *provide feedback*, and any other capabilities they would want. The thematic analysis was conducted by the authors through an iterative process, identifying 9 axial codes. The coding was conducted by the authors without independent raters; however, the themes were refined through multiple discussions to ensure consistency in interpretation. These codes, summarized in Figure 5, broadly represent four overarching themes:

- **Branching and Iteration:** highlights how participants interacted with the system during the querying process: **Branching queries** – 27 out of 50 participants found it useful to branch off a particular report to further investigate related queries. **Iterative questioning** – 2 out of 50 participants expressed a desire to iterate through a series of questions and answers to refine their understanding, giving them greater control.
- **Feedback Mechanisms:** reflects participants' desire for a system that evolves through user input: **Understanding how input is used** – 26 out of 50 participants emphasized the importance of transparency in how the system utilizes user feedback. **Providing input to improve** – 8 out of 50 participants wanted a feedback mechanism for correcting or guiding the system. **Evaluating based on historical data** – 3 out of 50 participants suggested using historical data for system evaluation.
- **Control Over Sources:** Participants desired control over brokerage reports or specific sources. Notably, 15 out of 50 participants highlighted the need to shortlist specific reports and trust certain sources, as evidenced by comments like, "I would personally trust some sources more than others". **Additional documents** – some participants (3/50) expressed desire to include documents such as annual reports or external sources.
- **Personalization and Adaptability:** Users expressed preferences for system customization: **Preferred sources** – 8 out of 50 participants had strong inclination toward specific analysts. **Generalize without bias** – 2 out of 50 participants wanted the system to generalize responses, and avoid over-reliance on their preferences. As one participant noted, "We need to see data from other sources in case we are missing a perspective."

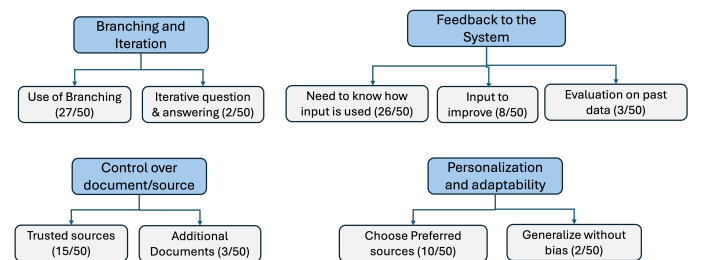


Figure 5: Codes from our qualitative analysis

## 5 Discussion

Our findings emphasize the critical interplay between *transparency* and *user control* in fostering trust in RAG systems. Text highlighting significantly improved participants' understanding of system reasoning and increased trust compared to confidence levels alone, consistent with prior research on explainability and traceability in AI systems [25]. Participants expressed a preference for greater control over source documents, such as filtering trusted reports and interacting with specific data. Personalization emerged as an important factor, enabling users to tailor the system to their preferences while avoiding cognitive overload or bias. These results highlight that transparency features must not only clarify system outputs but also provide actionable insights and foster user agency.

**Limitations.** While this study provides valuable insights, there are some limitations:

- **Task Order Effects:** The fixed task order may have introduced learning biases, favoring later conditions, such as text highlighting. Counterbalanced designs in future studies could mitigate this.
- **Domain-Specific Scope:** The focus on financial users and predefined equities limits generalizability to other domains or broader user groups.
- **Single-Session Study:** Trust was assessed in a single session, restricting insights into how trust evolves with repeated interactions. Longitudinal studies are needed to address this limitation.

## 6 Conclusion and Future Work

This study investigated the design and evaluation of a RAG-based assistant for equity research, focusing on transparency-enhancing features such as confidence indicators and source highlighting. While confidence levels provided useful context, source highlighting significantly improved user trust and understanding enabling traceability of system outputs. Participants also emphasized the importance of user control and personalization, such as filtering trusted reports and interacting with source data. These findings highlight the value of actionable transparency and user agency in high-stakes applications, offering guidance for designing trustworthy, user-centric RAG systems. Future work should explore longitudinal trust development, expand to other domains to enhance the applicability of these findings.

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